

EXPERIENCE

Graduation Project — Presentation Notes
Faculty of Computers and Informatics, Suez Canal University

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1. Project Overview

EXPERIENCE is a dual-platform system made to facilitate the tourism experience for people who are visiting Egypt for the first time, or for people who are willing to make their plan accurately based on what they prefer, or what they have experienced themselves in our simulation.

System Architecture

The system consists of two interconnected components:

- 3D Simulation (built in Unreal Engine 5)
- Mobile App (companion application)

The two components communicate bidirectionally through sync bridges.

2. The 3D Simulation

In our project we've made a 3D simulation for the experience of visiting the Giza Pyramids area. We built it as a third-person game, modeled as close to reality as we could, inside the Pyramids area that we've modeled as near as possible to real life.

Free Exploration

In the simulation the tourist can explore before they arrive in real life. First, they can walk freely in the area and see all three pyramids and the Sphinx.

Fast Travel System

We've made a fast travel system to simulate the internal transport buses in the Pyramids area. Wherever the player sees buses in the game, they can go there and fast travel through the area. The area is divided into five stations:

- 1. Sphinx Gate — the nearest gate

- 2. Parking
- 3. Khafre Pyramid
- 4. Khufu Pyramid
- 5. Menkaure Pyramid

The player can also walk through the Bazaar area near the Sphinx.

Main vs. Exploration Activities

All of the above are exploration activities that the player may explore freely without them counting as an official activity in the system.

3. Simulation Controls

Key	Action
WASD	Move the player
E	Ride / interact with the horse
N	Take a note (with screenshot)
L	View / select liked activities
M	View the map
ESC	Pause menu (access login)

4. Three Main Activities (Simulation–App Bridge)

Now we talk about the first bridge linking the simulation with the mobile app, which consists of three main activities:

- 1. Visit the Sphinx
- 2. Ride a horse
- 3. Enter the Great Pyramid of Khufu

Activity 1: Visit the Sphinx

The first activity is straightforward — the player simply visits the Sphinx area. No additional requirements beyond reaching the location.

Activity 2: Horse Ride

We've implemented a horse ride system. To try it, the player will need to go to the horses area, which is near the Sphinx area and the parking lot. The player can press M to view the map.

Once there, the player will find a sign. When they get close to it, a payment menu appears which they can use to pay to ride a horse. Once paid, the player can set around, get close to the standing horse, and ride it.

For the payment, the player should have money from the mobile app, which contains some mini fun games they can play to earn coins to spend in the simulation.

Activity 3: Enter the Great Pyramid of Khufu (Pyramid Interior)

The player will need to go to Khufu Pyramid and go up the stairs outside of it to trigger the payment menu, which allows them to go inside and enjoy reaching the King's Chamber.

Note: There is no much additional talk about this one as most details were covered in the horse ride section above — the payment flow works the same way.

5. Sync Bridges: Simulation ↔ Mobile App

These sync bridges make it easier for the user to remember what they've experienced in the simulation and take it with them when they actually go there in real life. The simulation was built inside Unreal Engine 5.

After Completing the Three Activities

After finishing the three activities, the player can press L to choose what they liked and send those choices to the mobile app.

Notes Feature (Press N)

At any time in the simulation, the player can press N to take a note with a picture of what they are currently seeing, and send it to the mobile app for future reference.

Account & Login

The player will first need to create an account from the mobile app, then log in to it from the desktop simulation by pressing ESC to open the pause menu and pressing the Log In button.

6. AI-Powered Tour Guide — Shrooq

The last major feature in the simulation is the AI-powered tour guide. The AI is made using ConvAI and her name is Shrooq Elkateeb. She is also the team leader of this project.

Placement in Simulation

The player will meet Shrooq at the entrance and beside each pyramid and the Sphinx.

How to Interact

- Get close to her and press Enter to text her — she responds with voice.
- Or hold T and talk to her through your mic whenever you are in the simulation.

Her Capabilities

She should understand and answer questions about:

- The project itself
- The Giza area and its history
- The simulation controls and what to do inside

She should NOT talk about technical implementation details unless explicitly asked.

7. The Mobile App

The mobile app should be a tourism hub for the user where they can manage their trips inside Egypt and plan them — starting from choosing the attractions, to date-adding, trip-booking, hiring a tour guide, and of course the simulation sync.

Key Features

- View and manage what was synced from the simulation (liked activities, notes with photos).
- Trip booking and tour guide selection.
- Mini fun games to earn coins spendable in the simulation.
- Chatbot about Egypt (general tourism information).

Sync from Simulation

The user can view what they chose from the simulation game, which helps them plan their real visit based on what they experienced virtually.

8. Team Members

#	Name	Role
1	Shrooq Elkateeb	Team Leader
2	Maged Ahmed	Unreal Engine Developer
3	Ahmed	Flutter Developer
4	Younna	Flutter Developer

5	Esraa	Blender / 3D Assets
6	Mina William	.NET Backend Developer
7	Mina Nabil	.NET Backend Developer
8	Shehd	UI/UX Designer

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